

Nanyang Technological University
SPMS/Division of Mathematical Sciences

2020/2021 Semester 1
TICS

MH2500 PROBABILITY AND INTRODUCTION TO STATIS-
Test 1

Name &
Matriculation Number

Score:

Mock Test

Mock Test

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1. (30 Points) What is the probability of getting a total of 6 or 10 when a pair of fair dices is tossed?

2. (40 Points)

Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement.

- (a) If we assume that at each draw each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red?
- (b) Now suppose that the balls have different weights, with each red ball having weight r and each white ball having weight w . Suppose that the probability that a given ball in the urn is the next one selected is its weight divided by the sum of the weights of all balls currently in the urn. Now what is the probability that both balls are red?

3. (30 Points) Let X be a Poisson random variable with parameter λ . Prove that

$$\mathbb{E}(X^n) = \lambda \mathbb{E}((X+1)^{n-1}).$$

Now use this result to show that $\mathbb{E}[X^3] = \lambda^3 + 3\lambda^2 + \lambda$.